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IMAGING APPARATUS, CONTROL METHOD, AND STORAGE MEDIUM

Abstract

An imaging apparatus includes at least one memory storing instructions, and at least one processor that, upon execution of the stored instructions cause the at least one processor to function as an acquisition unit configured to acquire a fluctuation amount of an image captured by an imaging unit, an image processing unit configured to perform correction processing for reducing fluctuation of the image based on the fluctuation amount acquired by the acquisition unit, and a determination unit configured to determine an exposure time of the imaging unit, wherein the determination unit determines the exposure time of the imaging unit based on at least one of the fluctuation amount acquired by the acquisition unit and a correction intensity of the correction processing.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an imaging apparatus, a control method, and a storage medium.

Description of the Related Art

[0002] For example, in a case where telephotography of a vessel or an aircraft is performed in a use case of a monitoring camera such as harbor surveillance and infrastructure monitoring, it is known that fluctuation of a subject image caused by uneven change of a refractive index of atmosphere (such as heat haze) deteriorates visibility of a subject. As a method of reducing such fluctuation, a method of smoothing an image in a time direction is known. However, in a case where a moving object is included in the image, the method of smoothing the image in the time direction may generate blur of the moving object. In particular, in the use case of the monitoring camera, it is desirable to perform automatic control so as to realize correction intensity suitable for a scene in order to cope with a phenomenon deteriorating the visibility of the image, such as fluctuation. As a method of correcting fluctuation other than the method of smoothing the image in the time direction, Japanese Patent Application Laid-Open No. 2015-177477 discusses a technique for correcting fluctuation by performing imaging in an exposure time and at a frame rate not affected by change of heat haze with the lapse of time.

SUMMARY OF THE INVENTION

[0003] According to an aspect of the present invention, an imaging apparatus includes at least one memory storing instructions, and at least one processor that, upon execution of the stored instructions cause the at least one processor to function as an acquisition unit configured to acquire a fluctuation amount of an image captured by an imaging unit, an image processing unit configured to perform correction processing for reducing fluctuation of the image based on the fluctuation amount acquired by the acquisition unit, and a determination unit configured to determine an exposure time of the imaging unit, wherein the determination unit determines the exposure time of the imaging unit based on at least one of the fluctuation amount acquired by the acquisition unit and a correction intensity of the correction processing.

[0004] Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1A and 1B are diagrams respectively illustrating an apparatus configuration and a functional configuration of an imaging apparatus.

[0006] FIG. 2 is a diagram illustrating fluctuation correction by smoothing in a time direction.

[0007] FIGS. 3A and 3B are diagrams illustrating comparison of temporal fluctuation at different shutter speeds.

[0008] FIGS. 4A and 4B are diagrams illustrating comparison of in-plane fluctuation at different shutter speeds.

[0009] FIGS. 5A, 5B, and 5C are diagrams respectively illustrating relationship between temporal

fluctuation and a shutter speed, relationship between in-plane fluctuation and the shutter speed, and relationship between the shutter speed and total fluctuation based on contributions of the temporal fluctuation and the in-plane fluctuation.

[0010] FIGS. **6A** and **6B** are diagrams respectively illustrating temporal change of fluctuation varied at a high frequency, and relationship between the fluctuation and the shutter speed.

[0011] FIG. **7** is a flowchart illustrating processing for determining execution/inexecution of the fluctuation correction.

[0012] FIG. **8** is a flowchart illustrating processing for setting a shutter speed for the fluctuation correction.

[0013] FIG. **9** is a diagram illustrating relationship between fluctuation and the shutter speed.

[0014] FIG. **10** is a flowchart illustrating processing for setting the shutter speed for the fluctuation correction.

[0015] FIG. **11** is a diagram illustrating relationship between fluctuation and the shutter speed.

DESCRIPTION OF THE EMBODIMENTS

[0016] Some embodiments of the present disclosure will be described below with reference to accompanying drawings. The embodiments described below are examples for implementing the present disclosure, and can be appropriately corrected or changed based on a configuration of an apparatus to which the present disclosure is applied and various conditions (e.g., use condition and use environment). Further, the embodiments described below may be partially combined, as appropriate. In the following embodiments, description will be given while the same components are denoted by the same reference numerals.

[0017] FIG. **1A** is a block diagram illustrating an apparatus configuration of an imaging apparatus according to a first embodiment. FIG. **1B** is a block diagram illustrating a functional configuration of the imaging apparatus according to the present embodiment.

(Apparatus Configuration)

[0018] The imaging apparatus according to the present embodiment includes an imaging optical system **101**, an imaging element **102**, a central processing unit (CPU) **103**, a random access memory (RAM) **104**, and a read only memory (ROM) **105**. The units are electrically connected via a bus **106**. The imaging optical system **101** is a lens connected to the imaging apparatus. Light having entered the imaging optical system **101** forms a subject image, and the imaging element **102** captures the subject image.

[0019] The imaging optical system **101** includes one or more lenses. The lenses include a lens subjected to coating for changing transmittance of infrared components, a lens subjected to special treatment for suppressing influence of aberration, and a lens that changes optical characteristics, such as a teleconverter (extender). Alternatively, the lenses include a lens that can be designed so as to have different functions for respective wavelengths, such as a meta-lens.

[0020] The imaging element **102** is a photoelectric conversion element that captures the subject image by the imaging optical system **101** and generates image data including a plurality of pixels or an image signal. The image data or the image signal includes information on a plurality of colors. The plurality of colors includes, for example, red, green, and blue. The image data is obtained by converting light having passed through color filters corresponding to the respective colors provided in front of the imaging element, into electric signals of a red signal, a green signal, and a blue signal by the imaging element. The color filters allow not only visible light components corresponding to the colors, red, green, and blue, but also a part of infrared components included in an invisible light region to pass therethrough. As the imaging element, a complementary metal oxide semiconductor (CMOS), a charge-coupled device (CCD), a single photon avalanche diode (SPAD), or the like is used.

[0021] The CPU **103** is a central processing unit totally controlling the imaging apparatus. The RAM **104** is a nonvolatile memory, and provides a work area to be used when the CPU **103** performs processing. In addition, the RAM **104** functions as a frame memory and a buffer memory.

The ROM **105** stores image data and programs for the CPU **103** to control the imaging apparatus. The ROM **105** may further store, for example, a threshold indicating intensity relating to fluctuation.

(Functional Configuration)

[0022] The imaging apparatus according to the present embodiment includes an acquisition unit **111**, a determination unit **112**, an exposure control unit **113**, and an image processing unit **114**. The functions are executed by software, and are implemented when programs stored in the ROM (storage unit) **105** are read to the RAM **104**, and the CPU **103** executes the programs.

[0023] The acquisition unit **111** acquires a fluctuation amount of an image captured by an imaging unit including the imaging optical system **101** and the imaging element **102**. Alternatively, the acquisition unit **111** may acquire information about fluctuation from a fluctuation correction value of the image processing unit **114**. The information about fluctuation includes, in addition to the fluctuation amount indicating a magnitude of fluctuation in the image, correction intensity during fluctuation correction.

[0024] In a case where the information about fluctuation is acquired from the acquired image, a difference of a pixel value of a pixel of interest among a plurality of consecutive images is calculated, and a magnitude of the calculated difference is acquired as the magnitude of the fluctuation. More specifically, a pixel at an edge part of a stationary subject in the image is regarded as the pixel of interest. In a case where the difference (change amount) is large in comparison of the pixel of interest among the plurality of consecutive images, the fluctuation amount is large, whereas in a case where the difference is small, the fluctuation amount is small. The acquisition unit **111** outputs the acquired fluctuation amount to the determination unit **112**.

[0025] The determination unit **112** determines a shutter speed (exposure time) based on the fluctuation amount acquired by the acquisition unit **111** and information from the exposure control unit **113**, thereby controlling the exposure time of the imaging element **102**. The shutter speed is determined so as to reduce fluctuation occurring in the image.

[0026] The exposure control unit **113** controls exposure by performing at least one of aperture driving control, shutter speed control, and gain control based on the fluctuation amount acquired by the acquisition unit **111** and the shutter speed determined by the determination unit **112**.

[0027] The image processing unit **114** performs image processing on the image data or the image signal input from the imaging element **102**. The image processing includes noise reduction processing, chroma and hue correction processing, gamma correction processing, and fluctuation correction processing by temporal smoothing. The image processing unit **114** performs correction processing for reducing the fluctuation of the image based on the fluctuation amount acquired by the acquisition unit **111**.

[0028] FIG. **2** is a diagram illustrating fluctuation correction by smoothing in a time direction such as simple moving average and weighted moving average. An image **201** is one frame of a moving image, and fluctuation has occurred in a building, which is a subject. An image **202** is a fluctuation-corrected image obtained by smoothing a plurality of frames. While fluctuation of the building has been reduced, blur has occurred in a bird, which is a moving object. An image **203** is a fluctuation-corrected image in a case where correction intensity of the smoothing processing in the time direction is weak compared with the image **202**. A method of changing the correction intensity described here is a method of changing the number of frames to be smoothed, but a method of changing a weight of the weighted moving average and the like may be used.

[0029] In the present embodiment, as in the case where the image **203** is generated, shutter speed control based on fluctuation is performed. As a result, fluctuation originally occurring in the image at imaging is reduced, and even when the correction intensity of the smoothing processing in the time direction is weakened, an image in which fluctuation is corrected and a moving object blur is suppressed can be obtained. A method of determining the shutter speed for reducing fluctuation will be described below.

[0030] First, the relationship between fluctuation and the shutter speed will be described with reference to FIGS. 3A and 3B and FIGS. 4A and 4B. The fluctuation is considered in terms of change of an amplitude with the lapse of time (hereinafter, temporal fluctuation) and in terms of change in a region of interest in a subject within one frame (hereinafter, in-plane fluctuation).

[0031] FIGS. 3A and 3B illustrate comparison of the temporal fluctuation at different shutter speeds. Dashed lines indicate random waveforms simulating fluctuation and indicate temporal changes of the fluctuation in one second. Solid lines in FIGS. 3A and 3B respectively indicate temporal fluctuation in a case where imaging is performed at a shutter speed of 1/30 seconds and 1/1000 seconds. Points on each of the solid lines indicate time points when one frame is generated. In this example, a frame rate is set to 60 fps. Therefore, 60 points are present in one second. The temporal fluctuation is calculated by averaging fluctuation (dotted line) by an accumulation time corresponding to the shutter speed before the time point when one frame is generated. The amplitude of the temporal fluctuation at time t in FIGS. 3A and 3B is greater at the shutter speed of 1/1000 seconds than at the shutter speed of 1/30 seconds. This is because an averaging period is reduced and responsiveness to change of fluctuation becomes high as the shutter speed is higher. FIG. 5A illustrates relationship between the shutter speed and the temporal fluctuation.

[0032] The temporal fluctuation is larger as the shutter speed is higher, and the temporal fluctuation is smaller as the shutter speed is lower.

[0033] FIGS. 4A and 4B illustrate comparison of the in-plane fluctuation at different shutter speeds. Dashed lines in FIGS. 4A and 4B also indicate temporal changes of the fluctuation as in FIGS. 3A and 3B. Solid lines in FIGS. 4A and 4B respectively indicate the in-plane fluctuation in the case where imaging is performed at the shutter speed of 1/30 seconds and 1/1000 seconds. The in-plane fluctuation is calculated by calculating difference between minimum fluctuation and maximum fluctuation (dotted lines) within the accumulation time corresponding to the shutter speed before the time point when one frame is generated. Cross marks in FIGS. 4A and 4B indicate minimum fluctuation and maximum fluctuation within the accumulation time at time t , and the in-plane fluctuation is smaller at the shutter speed of 1/1000 seconds than at the shutter speed of 1/30 seconds. This is because the accumulation time is reduced and the fluctuation change amount within the accumulation time is also reduced as the shutter speed is higher. FIG. 5B illustrates relationship between the shutter speed and the in-plane fluctuation. The in-plane fluctuation is smaller as the shutter speed is higher, and the in-plane fluctuation is larger as the shutter speed is lower.

[0034] FIG. 5C illustrates relationship between the shutter speed and total fluctuation based on contributions of the temporal fluctuation and the in-plane fluctuation. In the fluctuation simulated here, maximum fluctuation appears at a shutter speed **502**, and fluctuation at a shutter speed **501** lower than the shutter speed **502** and fluctuation at a shutter speed **503** higher than the shutter speed **502** are small. When influence of a moving object blur is considered, the moving object blur is likely to occur as the shutter speed becomes lower, and the moving object blur is less likely to occur as the shutter speed becomes higher. Therefore, comparing the shutter speed **501** and the shutter speed **503** at which the fluctuation is about the same, the shutter speed **503**, which is faster, is superior because the moving object blur can be suppressed and the fluctuation can be reduced. Further, at a shutter speed **504**, an effect of reducing the fluctuation greater than the effect at the shutter speed **503** can be achieved. However, the fluctuation is a phenomenon dependent on a temperature, a humidity, carbon dioxide concentration, and the like of atmosphere, and an amplitude and a frequency of the fluctuation depend on an environment where the fluctuation occurs. FIG. 6A illustrates a temporal change in one second, of the fluctuation that is varied at a frequency higher than a frequency of the fluctuation simulated in FIGS. 3A to 5C.

[0035] FIG. 6B illustrates relationship between the shutter speed and the total fluctuation based on contributions of the temporal fluctuation and the in-plane fluctuation in the fluctuation illustrated in FIG. 6A, and. As compared with FIG. 5C, a basic shape of the graph is not changed, but the shutter

speed at which the maximum fluctuation appears is shifted to a high-speed side, and an appropriate shutter speed depends on a frequency of the fluctuation. Therefore, the shutter speed for reducing the fluctuation is preferably set based on characteristics such as the amplitude and the frequency of the fluctuation.

[0036] There are some methods of setting the shutter speed based on the characteristics of the fluctuation. In the present embodiment, a method of setting the shutter speed by using characteristics that the moving object blur is small and the fluctuation is also small at the high shutter speed will be described. FIG. 7 is a flowchart illustrating main processing according to the present embodiment.

[0037] In step **S701**, execution/inexecution of the fluctuation correction is determined. Execution/inexecution of the fluctuation correction is determined based on user setting information or the fluctuation amount acquired by the acquisition unit **111**. In a case where the determination is performed based on the user setting information, setting information indicating whether to perform the fluctuation correction is stored in the RAM **104**, the ROM **105**, or an external storage device. Execution/inexecution of the fluctuation correction is determined based on the setting information acquired by the acquisition unit **111**. In a case where the determination is performed based on the acquired fluctuation amount, it is determined that the fluctuation correction is to be performed in a case where the acquired fluctuation amount is a predetermined amount or more. The predetermined amount can be optionally set by a user or a designer, and the fluctuation amount allowable by the user or the designer may be set. In a case where it is determined that the fluctuation correction is to be performed (YES in step **S701**), the processing proceeds to step **S702**. In a case where it is determined that the fluctuation correction is not to be performed (NO in step **S701**), the processing ends.

[0038] In step **S702**, the determination unit **112** determines the shutter speed based on the fluctuation amount. A determination method is described below.

[0039] In step **S703**, the image processing unit **114** performs the fluctuation correction by the image processing. Correction intensity of the fluctuation correction can be determined based on the fluctuation amount of an input image. Alternatively, the correction intensity can be determined by setting by the user. The correction intensity may be determined irrespective of a determination result of execution/inexecution of the fluctuation correction. In other words, the correction intensity of the fluctuation correction can be determined in any of steps **S701** to **S703** (it is sufficient to calculate correction intensity before processing for fluctuation correction is started). In a case where the fluctuation remains after step **S702**, the fluctuation correction by the image processing is performed at intensity corresponding to the remaining fluctuation. The fluctuation correction by the image processing is, for example, processing for smoothing the image in the time direction.

[0040] The processing in steps **S701** to **S703** may be performed again based on change of a scene and the like. For example, the processing in steps **S701** to **S703** can be performed again based on a determination condition, for example, a case where a change amount of the fluctuation amount of the input image is greater than or equal to a predetermined amount, a case where a change amount of exposure of the input image is greater than or equal to a predetermined amount, or a case where a predetermined time elapses after previous execution. That is, in a case where the change amount of exposure from proper exposure is greater than or equal to the predetermined amount, the exposure time may be redetermined or the exposure may be changed by changing a parameter relating to brightness different from the exposure time. In other words, in a case where the change amount of the fluctuation amount is less than the predetermined amount, the current exposure time may be maintained.

[0041] Next, processing for setting the shutter speed for the fluctuation correction (in step **S702**) is described with reference to a flowchart in FIG. 8.

[0042] In step **S801**, the determination unit **112** acquires a current shutter speed v_p from the exposure control unit **113**.

[0043] In step **S802**, the determination unit **112** acquires a fluctuation amount **q1** of the input image from the acquisition unit **111**.

[0044] In step **S803**, it is determined whether the fluctuation amount **q1** acquired in step **S802** is greater than or equal to a preset fluctuation allowable limit **qlim** (predetermined amount). The fluctuation allowable limit **qlim** is set to a value greater than or equal to the predetermined amount used for determining execution/inexecution of the fluctuation correction in step **S701**. The user or the designer may set the predetermined amount used for determining execution/inexecution of the fluctuation correction and the predetermined amount used for determining whether to control the shutter speed, to different amounts. As illustrated in FIG. 9, in a case where imaging is performed at the shutter speed of about 1/64 seconds, the fluctuation amount **q1** is greater than the fluctuation allowable limit **qlim**. In this case (YES in step **S803**), the processing proceeds to step **S804**. In a case where the fluctuation amount **q1** is less than the fluctuation allowable limit **qlim** (NO in step **S803**), the processing for setting the shutter speed for the fluctuation correction ends. The determination may be performed based on the correction intensity of the fluctuation correction in place of the fluctuation amount **q1**. More specifically, in step **S803**, it may be determined whether the correction intensity of the fluctuation correction is greater than or equal to predetermined intensity. In a case where a determination result is YES, the processing may proceed to step **S804**. Alternatively, the determination may be performed based on a combination of the fluctuation amount and the correction intensity. For example, the fluctuation amount **q1** (and correction intensity corresponding to fluctuation amount) and the correction intensity set by the user are compared, and a higher fluctuation amount or correction intensity may be used.

[0045] In step **S804**, the acquisition unit **111** acquires an exposure time range settable to the imaging unit, and the determination unit **112** sets, as the shutter speed for the fluctuation correction, a maximum shutter speed **vmax** (shortest exposure time) within the acquired exposure time range to the imaging unit. In the present embodiment, the shortest exposure time settable to the imaging unit is set, but the shutter speed is not necessarily the maximum shutter speed. It is sufficient to determine the shutter speed higher than the current shutter speed so as to reduce the fluctuation amount.

[0046] The exposure time range settable to the imaging unit indicates an exposure time range at which brightness of the image is not excessively lowered, or noise is not increased by a gain increased to maintain the brightness of the image. Therefore, the exposure time range settable to the imaging unit can be calculated by comprehensively considering parameters relating to the brightness of the image including luminance of the image, a noise amount, a shutter speed, a gain value, an aperture value, presence/absence of a neutral density (ND) filter, and the like. For example, allowable brightness or noise amount of the image is set in advance, and a shutter speed at which the brightness does not exceed the allowable brightness or the noise amount does not exceed the allowable noise amount in a case where the shutter speed is lowered is a lower limit of the settable exposure time range. As described above, it is sufficient to determine the shutter speed within a range where proper exposure is maintained. In step **S805**, the exposure control unit **113** controls the shutter speed to the shutter speed **vmax**.

[0047] In step **S806**, the determination unit **112** acquires a fluctuation amount **q2** at the shutter speed **vmax**, from the acquisition unit **111**.

[0048] In step **S807**, the determination unit **112** compares the fluctuation amounts before and after the setting of the shutter speed for the fluctuation correction. In FIG. 9, the fluctuation amount **q2** is less than the fluctuation amount **q1**, and the fluctuation is reduced after the setting of the shutter speed for the fluctuation amount is performed. As illustrated, in a case where the fluctuation amount after the setting of the shutter speed for the fluctuation correction is less than the fluctuation amount before the setting (YES in step **S807**), the processing for setting the shutter speed for the fluctuation correction ends.

[0049] In a case where the fluctuation amount is increased after the setting of the shutter speed for

the fluctuation correction is performed (NO in step **S807**), the determination unit **112** returns the shutter speed to the original shutter speed v_p in step **S808**. The exposure control unit **113** controls the shutter speed based on the shutter speed set by the determination unit **112**.

[0050] As described above, in the present embodiment, the exposure time of the imaging unit is determined based on at least one of the fluctuation correction amount acquired by the acquisition unit **111** and the correction intensity of the fluctuation correction. More specifically, different exposure times (shutter speeds) are determined based on whether the fluctuation correction amount (or correction intensity of fluctuation correction) is greater than or equal to a reference value. As a result, the shutter speed for reducing the fluctuation can be easily set, and the fluctuation correction can be performed while suppressing the moving object blur that is an adverse effect of the fluctuation correction by the image processing.

[0051] In the present embodiment, the case where the imaging unit including the imaging optical system **101** and the imaging element **102** is integral with the image processing unit **114** (corresponding to CPU **103**, RAM **104**, and ROM **105**) that performs the fluctuation correction is described. However, the imaging unit and the image processing unit **114** may be separated from each other. In other words, an external information processing apparatus may receive/acquire an image captured by the imaging apparatus via a network, and perform calculation of the fluctuation amount, calculation of the fluctuation correction amount, and determination of the shutter speed of the imaging apparatus based on the acquired image. In this case, the imaging apparatus can receive the determined shutter speed from the external information processing apparatus, and totally control the exposure including the gain based on the shutter speed received by the exposure control unit **113**.

[0052] In a second embodiment, not a case where relationship between the shutter speed and the fluctuation amount has a simple shape with one peak as illustrated in the first embodiment, but a case where minimum fluctuation appears at a certain shutter speed, for example, a case where the fluctuation has periodicity is considered. In this case, the optimum shutter speed can be set by setting the shutter speed by a method described below, rather than by simply making the shutter speed high. The processing for setting the shutter speed for the fluctuation correction (in step **S702**) according to the present embodiment will be described with reference to a flowchart in FIG. **10**. An apparatus configuration and a functional configuration of an imaging apparatus according to the present embodiment are similar to those according to the first embodiment. Therefore, description thereof will be omitted.

[0053] In step **S1001**, the determination unit **112** receives the current shutter speed from the exposure control unit **113**, and initializes the shutter speed at which the fluctuation amount becomes minimum (hereinafter, optimum shutter speed v_b) with the current shutter speed v_p .

[0054] In step **S1002**, the determination unit **112** acquires a current fluctuation amount q_p from the acquisition unit **111**, and initializes the minimum value of the fluctuation amount (hereinafter, minimum fluctuation amount q_{min}) varied by change of the shutter speed, with the current fluctuation amount q_p .

[0055] In step **S1003**, the determination unit **112** acquires the fluctuation amount while changing the shutter speed by a predetermined amount Δv , and determines the optimum shutter speed v_b at which the fluctuation amount becomes minimum (hereinafter, shutter scanning). At this time, a changing range of the shutter speed (hereinafter, shutter scanning range) is set. The shutter scanning range is set based on the current shutter speed v_p and the shutter speed range where the proper exposure calculated by the exposure control unit **113** is maintained. More specifically, a range between the shutter speed greater than or equal to the shutter speed before the shutter scanning and the shutter speed less than or equal to the maximum shutter speed within the proper exposure range is set. A shaded region illustrated in FIG. **11** is the shutter scanning range. The reason why a region on a side lower than the shutter speed before the shutter scanning is not subjected to the shutter scanning is to avoid occurrence of a moving image blur due to the shutter

speed for the fluctuation correction becoming lower than the originally-set shutter speed. In addition, when the exposure is maintained by the gain control during calculation of the shutter speed range where the proper exposure is maintained, noise in the image may be increased. Therefore, to prevent the noise from increasing, the shutter speed range where the proper exposure is maintained is desirably calculated while the gain is fixed.

[0056] In step **S1004**, the determination unit **112** sets a value that is increased by the predetermined amount Δv from the current shutter speed v_p , as a control target of the shutter speed, and outputs the set shutter speed to the exposure control unit **113**. The exposure control unit **113** controls the shutter speed based on the shutter speed received from the determination unit **112**.

[0057] In step **S1005**, the determination unit **112** acquires the current fluctuation amount q_p at the shutter speed set in step **S1004**, from the acquisition unit **111**.

[0058] In step **S1006**, the determination unit **112** determines whether the current fluctuation amount q_p is less than the minimum fluctuation amount q_{min} . In a case where the current fluctuation amount q_p is less than the minimum fluctuation amount q_{min} (YES in step **S1006**), the processing proceeds to step **S1007**. In a case where the current fluctuation amount q_p is greater than or equal to the minimum fluctuation amount q_{min} (NO in step **S1006**), subsequent steps **S1007** and **S1008** are skipped, and the shutter scanning is continued.

[0059] In step **S1007**, the determination unit **112** updates the optimum shutter speed v_b with the current shutter speed v_p .

[0060] In step **S1008**, the determination unit **112** updates the minimum fluctuation amount q_{min} with the current fluctuation amount q_p .

[0061] By repeating the processing in steps **S1004** to **S1008** until the current shutter speed v_p +the predetermined amount Δv goes out of the shutter scanning range, the shutter speed at which the fluctuation amount becomes minimum can be determined within the shutter scanning range. In FIG. **11**, the shutter speed at which the minimum fluctuation amount q_{min} is obtained within the shutter scanning range is the optimum shutter speed v_b .

[0062] In step **S1009**, the determination unit **112** sets the optimum shutter speed v_b as the control target of the shutter speed, and outputs the shutter speed to the exposure control unit **113**. The exposure control unit **113** controls the shutter speed based on the shutter speed received from the determination unit **112**. As described above, the shutter speed at which the fluctuation amount becomes minimum is determined by scanning. Thus, the shutter speed can be controlled to the shutter speed at which the fluctuation amount is reduced with high accuracy as compared with the first embodiment, and a higher fluctuation reduction effect can be achieved.

Modification

[0063] As a modification of the present embodiment, it may be determined in step **S1006** whether the acquired current fluctuation amount q_p is less than or equal to a predetermined amount, and in a case where the acquired current fluctuation amount q_p is less than or equal to the predetermined amount, the shutter scanning may be terminated. As a result, the processing in the flowchart can be terminated when the fluctuation amount becomes less than or equal to a desired fluctuation amount, without performing the shutter scanning until the fluctuation amount becomes minimum. Thus, effects superior to the effects obtained by the present embodiment in terms of a processing speed and processing efficiency can be achieved.

OTHER EMBODIMENTS

[0064] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for

example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0065] While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments but is defined by the scope of the following claims.

[0066] This application claims the benefit of Japanese Patent Application No. 2024-034867, filed Mar. 7, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An imaging apparatus comprising: at least one memory storing instructions; and at least one processor that, upon execution of the stored instructions cause the at least one processor to function as: an acquisition unit configured to acquire a fluctuation amount of an image captured by an imaging unit; an image processing unit configured to perform correction processing for reducing fluctuation of the image based on the fluctuation amount acquired by the acquisition unit; and a determination unit configured to determine an exposure time of the imaging unit, wherein the determination unit determines the exposure time of the imaging unit based on at least one of the fluctuation amount acquired by the acquisition unit and a correction intensity of the correction processing.
2. The imaging apparatus according to claim 1, wherein the determination unit determines the exposure time to reduce fluctuation occurring in the image, based on at least one of the fluctuation amount acquired by the acquisition unit and the correction intensity of the correction processing.
3. The imaging apparatus according to claim 1, wherein, in a case where it is determined that the fluctuation amount acquired by the acquisition unit is greater than or equal to a predetermined amount, the determination unit determines an exposure time shorter than a current exposure time of the imaging unit, as the exposure time of the imaging unit.
4. The imaging apparatus according to claim 3, wherein the acquisition unit acquires an exposure time range settable to the imaging unit, and wherein, in a case where it is determined that the fluctuation amount acquired by the acquisition unit is greater than or equal to the predetermined amount, the determination unit determines a shortest exposure time within the exposure time range settable to the imaging unit, as the exposure time of the imaging unit.
5. The imaging apparatus according to claim 1, wherein the acquisition unit acquires a plurality of fluctuation amounts of a plurality of images captured by the imaging unit while changing the exposure time of the imaging unit by a predetermined amount, and wherein the determination unit determines an exposure time at which fluctuation occurring in the images captured by the imaging unit is less than a predetermined amount, as the exposure time of the imaging unit, based on the plurality of fluctuation amounts acquired by the acquisition unit.
6. The imaging apparatus according to claim 1, wherein the determination unit determines the exposure time of the imaging unit within a range where proper exposure of the image captured by the imaging unit is maintained.
7. The imaging apparatus according to claim 1, wherein the determination unit determines the

exposure time of the imaging unit within a range where a gain of an imaging element included in the imaging unit is maintained.

8. The imaging apparatus according to claim 1, wherein the acquisition unit acquires the fluctuation amount at a predetermined time interval, wherein, in a case where a change amount of a current fluctuation amount is less than a predetermined amount, the fluctuation amount acquired when a current exposure time is determined being a reference, the determination unit maintains the current exposure time, and wherein, in a case where the change amount of the current fluctuation amount is greater than or equal to the predetermined amount, the determination unit redetermines the exposure time.

9. The imaging apparatus according to claim 1, wherein, in a case where a change amount of exposure from proper exposure is greater than or equal to a predetermined amount, the determination unit redetermines the exposure time or changes exposure by changing a parameter relating to brightness different from the exposure time.

10. The imaging apparatus according to claim 1, wherein, in a case where a predetermined time elapses, the determination unit redetermines the exposure time.

11. The imaging apparatus according to claim 1, wherein the fluctuation amount is calculated based on a difference of a pixel value among a plurality of images which are frames of a moving image captured by the imaging unit.

12. The imaging apparatus according to claim 11, wherein the correction processing is temporal smoothing of the images.

13. A control method comprising: acquiring a fluctuation amount of an image captured by an imaging unit; performing correction processing for reducing fluctuation of the image based on the acquired fluctuation amount; and determining an exposure time of the imaging unit, wherein the exposure time of the imaging unit is determined based on the acquired fluctuation amount or a correction intensity of the correction processing.

14. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a method comprising: acquiring a fluctuation amount of an image captured by an imaging unit; performing correction processing for reducing fluctuation of the image based on the acquired fluctuation amount; and determining an exposure time of the imaging unit, wherein the exposure time of the imaging unit is determined based on the acquired fluctuation amount or a correction intensity of the correction processing.
